Biometrics, 2025, 81(4), ujaf162
https://doi.org/10.1093/biomtc/ujaf162
Biometric Methodology

OXFORD

Estimating heterogeneous treatment effects for general responses

Zijun Gao [1,*] and Trevor Hastie[2]

[1]Department of Data Sciences and Operations, Marshall Business School, University of Southern California, Los Angeles, CA 90089, United States, [2]Department of Statistics and Department of Biomedical Data Science, Stanford University, Stanford, CA 94305, United States

*Corresponding author: Zijun Gao, Department of Data Sciences and Operations, Marshall Business School, University of Southern California, Los Angeles, CA 90089, USA (zijungao@marshall.usc.edu).

ABSTRACT

Heterogeneous treatment effect models allow us to compare treatments at subgroup levels and are becoming increasingly popular in applications such as personalized medicine, advertising, and education. Regardless of the type of responses (continuous, binary, count, survival), most causal estimands focus on the differences between the treatment and control conditional means. In this paper, we propose an alternative estimand, DINA—the DIfference in NAtural parameters—to quantify heterogeneous treatment effects motivated by exponential families and the Cox model. Despite the type of responses, DINA is both convenient and more practical for modeling the influence of covariates on the treatment effect. Additionally, we introduce a meta-algorithm for DINA, enabling practitioners to utilize powerful off-the-shelf machine learning tools for the estimation of nuisance functions. This meta-algorithm is also statistically robust to errors in the nuisance function estimation. We demonstrate the efficacy of our method in combination with various machine learning base-learners on both simulated and real datasets.

KEYWORDS: causal inference; Cox model; exponential family; heterogeneous treatment effect; Neyman orthogonal score.

1 INTRODUCTION

The potential outcome model (Rubin, 1974) has received wide attention (Imbens and Rubin, 2015; Rosenbaum et al., 2010) in the field of causal inference. Recently, there has been a focus on the estimation of heterogeneous treatment effects (HTE), which allows the estimated effect to depend on subject-specific features. Applications include personalized medicine (Lesko, 2007; Low et al., 2016), personalized education (Murphy et al., 2016), and personalized advertisement (Bennett and Lanning, 2007).

For continuous responses, the difference in conditional means, often referred to as the conditional average treatment effect (CATE), is a commonly-used estimand of HTE (Powers et al., 2018; Wendling et al., 2018). However, for binary responses or count data, no consensus of the estimand has been reached, and a variety of quantities have been considered. For instance, for dichotomous responses, conditional success probability differences, conditional success probability ratios, and conditional odds ratios have all been employed (Imbens and Rubin, 2015; Tian et al., 2014). In this paper, we propose to estimate a unified quantity—the DIfference in NAtural parameters (DINA), motivated by exponential families and the Cox model (Cox, 1972). DINA is the conditional mean difference for continuous responses, the conditional log-odds ratio for binary responses, the conditional log mean ratio for count responses, and the conditional log hazard ratio for survival outcomes.

DINA is more appealing for general responses compared to CATE for reasons below:

(1) In practical applications, comparisons on the natural parameter scale are frequently used.
For binary responses, the log of the conditional odds ratio is a frequently used metric in clinical trials for assessing diagnostic performance (Glas et al., 2003; Rothman, 2012), and is approximately equivalent to the relative risk for rare diseases. In the Cox model, the log of the hazard ratio, which measures the likelihood of an event occurring in the treatment arm relative to the control arm, is commonly reported.
(2) It is convenient to model the influence of covariates to the treatment effect on the natural parameter scale. For binary responses or count data, the types of outcomes impose implicit constraints, such as zero-one or non-negative values, making it difficult to model covariates' impact on the outcome scale. In contrast, natural parameters are supported on the real line, which easily accommodate various types of covariate dependence.
(3) The difference in conditional means can display uninteresting or over-complicated heterogeneity due to

Received: June 28, 2024; Revised: September 27, 2025; Accepted: October 9, 2025
© The Author(s) 2025. Published by Oxford University Press on behalf of The International Biometric Society. This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (https://creativecommons.org/licenses/by-nc/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site-for further information please contact journals.permissions@oup.com

Downloaded from https://academic.oup.com/biometrics/article/81/4/ujaf162/8403946 by guest on 07 June 2026

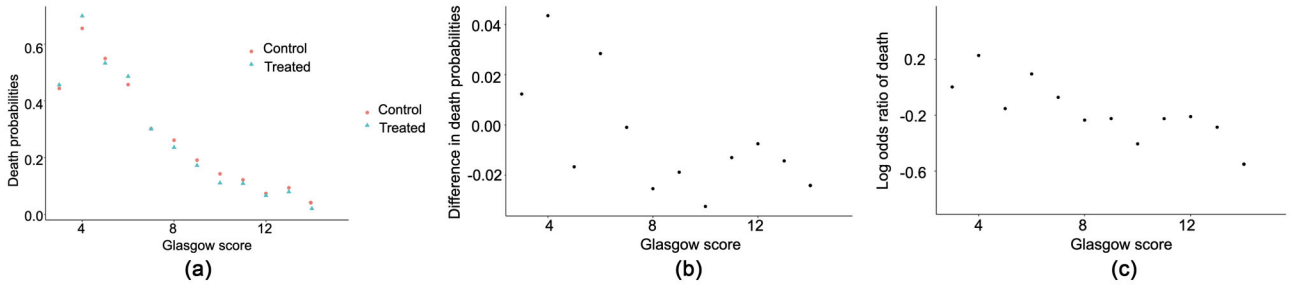


FIGURE 1 Treatment effect of drug Tranexamic acid and Glasgow score. Panel (a) displays the probabilities of death for different Glasgow scores. Panel (b) displays the difference of the treatment and control death probabilities. Panel (c) displays the log-odds ratio of the treatment and control death probabilities.

the support constraints. For instance, consider a scenario where the disease risk without a vaccine is 10% among older people and 1% among young people. Given the substantial disparity in room for improvement between age groups (10% versus 1%), the absolute risk difference is unlikely to be identical; in contrast, the relative risk may remain constant across different ages. Another example is the CRASH-3 data (Dewan et al., 2012) examining how the effect of Tranexamic acid on death rates varies with Glasgow scores (Figure 1). The differences in death probabilities (panel b) exhibit a complex nonlinear dependence on the Glasgow score, whereas the difference in the log-odds ratio of death (panel c) shows a simpler linear relationship.

For DINA, we propose an estimator motivated by the celebrated Robinson's method (Robinson, 1988) for CATE and its extension the R-learner (Nie and Wager, 2021). Our method consists of two steps:

(1) Estimate nuisance functions.
(2) Estimate DINA with nuisance function estimators plugged in.

The method is designed to be locally insensitive to incorrect nuisance function estimators. By separating the estimation of nuisance functions from that of DINA, we can use powerful machine-learning tools for the estimation of nuisance functions.

In Section 2, we formulate the problem and discuss related works. In Section 3, we discuss DINA estimator for exponential families. In Section 4, we extend the DINA estimator to the Cox model. In Sections 5 and 6, we apply the DINA estimator to simulated data and the SPRINT dataset. We conclude this paper with discussions in Section 7. Literature review, proofs, and method extensions are provided in the Supplementary Materials.

2 BACKGROUND

2.1 Problem formulation

We adopt the Neyman–Rubin potential outcome model. Each unit is associated with a covariate vector $X \in \mathbb{R}^d$, a treatment assignment indicator $W \in \{0, 1\}$, and two potential outcomes $Y(0)$, $Y(1) \in \mathbb{R}$. We observe the response $Y = Y(1)$ if the unit is under treatment, i.e., $W = 1$, and $Y = Y(0)$ if the unit is under control, i.e., $W = 0$.

Suppose we have access to n units drawn i.i.d. from a super-population model,

$$X \overset{\text{i.i.d.}}{\sim} \mathbb{P}_X, \tag{1}$$

$$W \mid X = x \overset{\text{ind.}}{\sim} \text{Ber}(e(x)), \tag{2}$$

$$\begin{cases} Y(1) \mid X = x \sim \mathbb{P}_1(\cdot \mid x), \\ Y(0) \mid X = x \sim \mathbb{P}_0(\cdot \mid x), \end{cases} \tag{3}$$

where $\mathbb{P}_X$ denotes the distribution of covariates, $e(x)$ denotes the propensity score, and $\mathbb{P}_1$ and $\mathbb{P}_0$ denote the distributions of potential outcomes.

2.1.1 Exponential family

We aim to estimate the heterogeneous treatment effects, denoted by $\tau(x)$, where the exact form of $\tau(x)$ depends on the response type.

(1) For continuous data, we use the difference of the conditional means

$$\tau(x) := \mathbb{E}[Y(1) \mid X = x] - \mathbb{E}[Y(0) \mid X = x]. \tag{4}$$

(2) For binary responses, we use the difference of log conditional odds, which is equivalent to the log of conditional odds ratio,

$$\tau(x) := \log\left(\frac{\mathbb{P}(Y(1) = 1 \mid X = x)}{\mathbb{P}(Y(1) = 0 \mid X = x)}\right) - \log\left(\frac{\mathbb{P}(Y(0) = 1 \mid X = x)}{\mathbb{P}(Y(0) = 0 \mid X = x)}\right). \tag{5}$$

(3) For count data, we use the difference of the log of conditional means, i.e., the log of conditional mean ratio,

$$\tau(x) := \log\left(\mathbb{E}[Y(1) \mid X = x]\right) - \log\left(\mathbb{E}[Y(0) \mid X = x]\right). \tag{6}$$

Downloaded from https://academic.oup.com/biometrics/article/81/4/ujaf162/8403946 by guest on 07 June 2026

If the responses follow Gaussian, Bernoulli, or Poisson distributions, the estimands above correspond to the DINA functions of treatment and control groups, which gives rise to the name "DINA." When the responses do not follow these distributions, the quantities remain well-defined and meaningful (Section 3).

2.1.2 Cox model

For time-to-event responses, we let $\lambda_0(y\,x)$ be the conditional control hazard rate at time y, and similarly we define $\lambda_1(y\,x)$. We target the difference of the log of conditional hazards, i.e., the log of hazards ratio,

$$\tau(y\,x) := \log\left(\lambda_1(y\,x)\right) - \log\left(\lambda_0(y\,x)\right) \tag{7}$$

We follow the Cox model (Cox, 1972) and make the proportional hazards assumption.

Assumption 1 (Proportional hazards): *The hazard rate functions follow*

$$\lambda_0(y\,x) = \lambda(y)e^{\eta_0(x)}, \quad \lambda_1(y\,x) = \lambda(y)e^{\eta_1(x)},$$

where $\lambda(y)$ denotes the baseline hazard function and $e^{\eta_0(x)}$, $e^{\eta_1(x)}$ denote the exponential tilting functions.

Under Assumption 1, $\tau(y\,x)$ does not depend on y. Below we will omit y and use $\tau(x)$.

Survival responses usually come with censorship. We let $C(0)$, $C(1) \in [0, \infty]$ be the counterfactual censoring times and C be the observed censoring time where $C = C(1)$ if $W = 1$ and $C = C(0)$ if $W = 0$. Let $\Delta := \mathbb{1}_{\{C \geq Y\}}$ be the observed censoring indicator. The observed responses are pairs $(Y^c := \min\{Y, C\}, \Delta)$. We make the following assumption on the counterfactual censoring times:

Assumption 2 (Censoring mechanism): *The counterfactual censoring times are independent of the survival times, given the covariates and the treatment assignment,*

$$Y(0) \perp\!\!\!\perp C(0) \mid W, X,$$

$$Y(1) \perp\!\!\!\perp C(1) \mid W, X,$$

and the counterfactual censoring times are unconfounded,

$$C(1), C(0) \perp\!\!\!\perp W \mid X.$$

Assumption 2 implies that there are no unmeasured factors influencing the counterfactual censoring times and survival times. The assumption also assumes away confounders beyond the covariates X that could affect both the censoring times and the treatment assignment.

2.2 Robinson's method and R-learner

In Section 1 of Supplementary Materials, we discuss related works. Below we focus on Robinson's method (Robinson, 1988) and the R-learner (Nie and Wager, 2021) for CATE (we use two method names interchangeably below), which motivate our procedure. To illustrate, consider the additive error model

$$Y(1) = \eta_1(X) + \varepsilon, \quad Y(0) = \eta_0(X) + \varepsilon, \tag{8}$$

where $\eta_0(x)$, $\eta_1(x)$ denote the conditional control and treatment group mean functions, and the error ε satisfies $\mathbb{E}[\varepsilon \mid X] = 0$. The estimand CATE is the difference in conditional means $\tau(x) = \eta_1(x) - \eta_0(x)$. Let

$$m(x) := \mathbb{E}[Y \mid X = x] = \{1 - e(x)\}\eta_0(x) + e(x)\eta_1(x) \tag{9}$$

be the marginal mean function, then model (8) can be reparametrized as

$$Y = \eta_0(X) + W\tau(X) + \varepsilon = m(X) + \{W - e(X)\}\tau(X) + \varepsilon. \tag{10}$$

Based on the reparametrization (10), Robinson (1988) further assumes $\tau(x) = x^\top\beta$ and proposes the two-step procedure:

(1) Estimation of nuisance functions. Estimate the propensity score $e(x)$ and the marginal mean function $m(x)$.
(2) Least squares. Substitute the nuisance-function estimators $\hat{e}(x)$, $\hat{m}(x)$ from Step 1 in (10) and solve for β by least squares.

The R-learner adopts the same reparametrization (10) and the two-step procedure, but estimates $\tau(x)$ nonparametrically. Robinson's method and the R-learner use black-box predictors for nuisance function estimation and are only mildly sensitive to inaccurate nuisance function estimators $\hat{m}(x)$, $\hat{e}(x)$. In this paper, we extend the R-learner to DINA.

3 DINA FOR EXPONENTIAL FAMILY

In this section, we introduce our DINA estimator suitable for responses generated from exponential family. For illustration, we begin with a working model for the potential outcomes to motivate the proposed method, and defer discussion of more general cases to the end of the section. The working model is given by

$$\begin{cases} f_1(y \mid X = x) = \kappa(y)e^{y\eta_1(x) - \psi(\eta_1(x))}, \\ f_0(y \mid X = x) = \kappa(y)e^{y\eta_0(x) - \psi(\eta_0(x))}. \end{cases} \tag{11}$$

Here f_0, f_1 denote the densities of the control group and the treatment group respectively, $\eta_0(x)$, $\eta_1(x)$ denote the associated natural parameter functions, $\psi(\eta)$ is the cumulant generating function, and $\kappa(y)$ is the carrier density. Our estimand DINA is $\tau(x) = \eta_1(x) - \eta_0(x)$.

Analogous to Eq. (9), we define the baseline $\nu(x)$ as a mixture of natural parameter functions weighted by a modified propensity score $a(x)$,

$$\begin{aligned} \nu(x) :=& \quad a(x)\eta_1(x) + \{1 - a(x)\}\eta_0(x), \\ a(x) :=& \quad \frac{e(x)V_1(x)}{e(x)V_1(x) + \{1 - e(x)\}V_0(x)}. \end{aligned} \tag{12}$$

Here $V_1(x) = \frac{d\mu}{d\eta}(\eta_1(x))$ and $V_0(x) = \frac{d\mu}{d\eta}(\eta_0(x))$ are the variance functions for the exponential family, $\mu(\cdot)$ being the inverse of the canonical link function. This allows us to

Downloaded from https://academic.oup.com/biometrics/article/81/4/ujaf162/8403946 by guest on 07 June 2026

reparametrize $\eta_0(x)$, $\eta_1(x)$ as

$$\eta_w(x) = \nu(x) + \{w - a(x)\}\tau(x), \quad w \in \{0, 1\}. \tag{13}$$

The R-learner for the Gaussian distribution uses $a(x) = e(x)$, $\nu(x) = \mathbb{E}[Y \mid X = x]$.

Based on (13), we propose the following two-step estimator (details in Algorithm 1):

(1) Estimation of nuisance functions. We estimate $a(x)$ and $\nu(x)$ in (12), using estimators of the propensity score $e(x)$ and the natural parameter functions $\eta_0(x)$ and $\eta_1(x)$.
(2) Maximum likelihood estimator (MLE). We maximize the likelihood of $\tau(x)$ corresponding to (13) with nuisance-function estimators $\hat{a}(x)$, $\hat{\nu}(x)$ from Step 1 plugged in,

$$\max_{\tau'(x)} \ell(\tau'(x)\,\hat{a}(x), \hat{\nu}(x)) := \frac{1}{n}\sum_{i=1}^{n} Y_i\,(\hat{\nu}(X_i) + \{W_i - \hat{a}(X_i)\}\tau'(X_i))$$
$$-\psi\,(\hat{\nu}(X_i) + \{W_i - \hat{a}(X_i)\}\tau'(X_i)) \tag{14}$$

The method described above extends beyond the working model (13). More generally, based on the type of response variable, we choose a link function g (analogous to $(\psi')^{-1}$ in the exponential family (13)) to transform the outcome from the original scale to the "natural parameter" scale. Next, we define the baseline functions,

$$\eta_w(x) = g\,(\mathbb{E}[Y(w) \mid X = x])\,, \quad w \in \{0, 1\}, \tag{15}$$

which reduces to the natural parameter functions under (13) with $g = (\psi')^{-1}$. When the exponential family working model is misspecified, the proposed $\hat{\tau}(x)$ still estimates the quantity $\eta_1(x) - \eta_0(x)$, i.e., the difference in transformed conditional expectations between the treatment and control groups, and Proposition 1, which will be introduced in Section 3.3, still holds. We also show that our proposed nuisance functions (12) lead to the Neyman orthogonal score derived from the concentration-out approach (Newey, 1994). Details are provided in the Supplementary Materials Section 3.1.1.

3.1 Interpretation of nuisance functions

We provide insights into Algorithm 1 by comparing it to the R-learner. As discussed above, Algorithm 1 is an extension of the R-learner, and shares its motivation as well as the method framework. The major difference lies in the nuisance functions (12). In the R-learner, $a(x)$ equals the propensity function $e(x)$; in the proposed DINA estimator, $a(x)$ not only depends on $e(x)$ but also the variances $V_0(x)$ and $V_1(x)$. By the decomposition,

$$\log\frac{a(x)}{1-a(x)} = \log\frac{e(x)}{1-e(x)} + \log\frac{V_1(x)}{V_0(x)}. \tag{16}$$

$a(x)$ is large if a unit is likely to be treated or their response under treatment has higher variance than under control. As a result, $a(x)$ not only balances the sizes of the treatment and control groups but also mitigates the variance differences between them. For more comments on the adjustment based on $V_w(x)$, see the multiple-level treatment setting in the Supplementary Materials Section 2.

3.2 Algorithm

Similar to the R-learner, we estimate the DINA in (13) in two steps. We estimate nuisance functions (12) and the DINA using independent subsamples so that the estimation of the nuisance functions does not interfere with that of DINA. We employ cross-fitting (Chernozhukov et al., 2018) to restore data efficiency. We now give details.

In the first step, we estimate the nuisance functions $\nu(x)$ and $a(x)$ in (12). These in turn depend on the propensity score $e(x)$, and separate estimators of $\eta_0(x)$ and $\eta_1(x)$. The propensity scores can be estimated by any classification method that provides probability estimates. The functions $\eta_0(x)$ and $\eta_1(x)$ can be separately estimated by any suitable method that operates within the particular exponential family. The variance functions $V_0(x)$ and $V_1(x)$ are functions of $\eta_0(x)$ and $\eta_1(x)$, and can be estimated by inserting $\hat{\eta}_0(x)$ and $\hat{\eta}_1(x)$ into the corresponding expressions. Finally, we plug the nuisance estimators into (12) and get $\hat{a}(x)$ and $\hat{\nu}(x)$. Note that in the case of the Gaussian distribution, $a(x) = e(x)$, $\nu(x) = \mathbb{E}[Y \mid X = x]$, and we can directly estimate $\nu(x)$ by any conditional mean estimator of Y given X and avoid having to separately estimate $\hat{\eta}_0(x)$ and $\hat{\eta}_1(x)$.

In the second step, we maximize the log-likelihood corresponding to (13) with the estimators $\hat{\nu}(x)$ and $\hat{a}(x)$ from the first step. The function $\nu(x)$ can be regarded as an *offset*, and $\tau(x)$ can be considered as the varying coefficient of $W - a(X)$. The second step can be implemented, for example, using the `glm` function in R with the additional assumption that $\tau(x)$ is linear in x.

From the practical perspective, Algorithm 1 decouples the estimation of DINA from that of the nuisance functions and allows for flexible estimation of nuisance functions. Though the nuisance functions $a(x)$ and $\nu(x)$ are complicated, various methods are applicable since there are no missing data in the nuisance-function estimation. By modularization, in step 1, we can use any off-the-shelf methods:

3.3 Theoretical properties

The motivation behind the method is to gain robustness to nuisance functions, and we make our idea rigorous in the following proposition. In the proposition, we replace $\hat{\beta}$ with β_n to explicitly indicate its dependence on the sample size. For a function $g(x)$, $\|g(x)\|_2$ is defined as $(\mathbb{E}[g^2(X)])^{1/2}$.

Proposition 1: *For η_w in Eq. (15), under conditions:*

(1) Treatment effect is a linear function of x, that is, $\tau(x) = x^\top\beta$;
(2) Covariates X are bounded, the true parameter β is in a bounded region $\mathcal{B}$, nuisance functions $a(x)$, $\nu(x)$ and nuisance-function estimators $a_n(x)$, $\nu_n(x)$ are uniformly bounded;
(3) The minimal eigenvalues of the score derivative $\nabla_\beta s(a(x), \nu(x), \beta)$ in $\mathcal{B}$ are lower bounded by C, $C > 0$.

Downloaded from https://academic.oup.com/biometrics/article/81/4/ujaf162/8403946 by guest on 07 June 2026

We split the data randomly equally into two folds.1. On fold one, we obtain nuisance-function estimators $\hat{a}(x)$ and $\hat{\nu}(x)$.

(1) Estimation of $e(x)$.
(2) Estimation of $\eta_0(x)$,$\eta_1(x)$. Estimate the natural parameterfunctions $\eta_0(x)$ and $\eta_1(x)$ based on thecontrol group and the treatment group respectively,such as through fitting generalized linear models,random forests, or boosting.
(3) Substitution. Plug the estimators $\hat{e}(x)$, $\hat{\eta}_0(x)$, $\hat{\eta}_1(x)$ into **??** and get $\hat{a}(x)$. Let $\hat{\nu}(x) = \hat{a}(x)\hat{\eta}_1(x) + \{1 - \hat{a}(x)\}\hat{\eta}_0(x)$.

2. On fold two, we obtain $\hat{\tau}(x)$ by maximizing thelog-likelihood (Eq.~**??**).with $\hat{a}(x)$ and $\hat{\nu}(x)$ plugged in.
3. We swap the two folds, obtain another estimate, and output the average of the estimators.

Algorithm 1: Exponential family

If $\|a_n(x) - a(x)\|_2$, $\|\nu_n(x) - \nu(x)\|_2 = O(c_n)$, $c_n \to 0$, *then*

$$\|\beta_n - \beta\|_2 = \tilde{O}\left(c_n^2 + n^{-1/2}\right).$$

The estimator is efficient if the likelihood (11) is correctly specified and $c_n = o(n^{-1/4})$.

The linearity assumption $\tau(x) = x^\top\beta$ can be replaced by $\tau(x) = z(x)^\top\beta$, for any known functions of the covariates $z(x)$. For instance, if the treatment effect is believed to be homogeneous, we use $z(x) = 1$; if we are interested in subgroup treatment effects, we can design group indicators to specify the desired subgroups. If the set of basis is sufficiently expressive, then the linear model can approximate smooth functions reasonably well.

Proposition 1 states that for $\hat{\tau}(x) = x^\top\hat{\beta}$ to achieve a certain accuracy, the conditions on $\hat{a}(x)$ and $\hat{\nu}(x)$ are relatively weak. This implies the method is locally insensitive to the nuisance functions, and thus is robust to noisy plugged-in nuisance-function estimators. In other words, Algorithm 1 can be viewed as an accelerator: given crude estimators $\hat{\eta}_1(x)$, $\hat{\eta}_0(x)$, and $\hat{e}(x)$ as inputs, it produces DINA estimates that are twice as accurate as the naive difference $\hat{\eta}_1(x) - \hat{\eta}_0(x)$. As a corollary of Proposition 1, if both $\hat{a}(x)$ and $\hat{\nu}(x)$ are estimated at rate $n^{-1/4}$, the resulting DINA estimator achieves the parametric rate $n^{-1/2}$.

Proposition 1 remains valid for general η_w as defined in Eq. (15), even when the exponential family model (11) is misspecified. In fact, when the nuisance functions are fit using generalized linear models or machine learning methods such as XGBoost with the negative log-likelihood as the loss, $a(x)$ and $\nu(x)$ converge to the "best-fitting" values that minimize the Kullback–Leibler divergence between the true data-generating process and the working model even if misspecified. However, correct model specification is required to establish the efficiency result.

4 DINA FOR COX MODEL

Let $\Lambda(y)$ be the baseline cumulative hazard function, i.e., $\Lambda(y) = \int_0^y \lambda(t)dt$. We assume

$$\begin{cases} \mathbb{P}(Y(1) \geq y \mid X = x) = e^{-\Lambda(y)e^{\eta_1(x)}}, \\ \mathbb{P}(Y(0) \geq y \mid X = x) = e^{-\Lambda(y)e^{\eta_0(x)}}, \end{cases} \tag{17}$$

where the baseline hazard function $\Lambda(y)$ is shared by all subjects, and the hazard of a subject is the baseline multiplied by a tilting function independent of the survival time.

Analogous to Eq. (12), we define the baseline $\nu(x)$, the modified propensity score $a(x)$

$$a(x) = \frac{e(x)\mathbb{P}(C \geq Y \mid W = 1, X = x)}{e(x)\mathbb{P}(C \geq Y \mid W = 1, X = x) + \{1 - e(x)\}\mathbb{P}(C \geq Y \mid W = 0, X = x)}, \tag{18}$$

$$\nu(x) = a(x)\eta_1(x) + \{1 - a(x)\}\eta_0(x).$$

The value of $a(x)$ is large when the propensity score is high, or when the ratio of not-censored probabilities between the treatment group and the control group is large. As shown in the Supplementary Materials Section 2, if the baseline hazard is known, then plugging the above $a(x)$, $\nu(x)$ into Algorithm 1 leads to the hazard ratio estimator based on the full likelihood, which is robust to the nuisance estimators $\hat{a}(x)$, $\hat{\nu}(x)$ (Proposition 1).

Instead of focusing on the full likelihood, Cox (1972) proposes to maximize the partial likelihood. The partial likelihood is prevalent in practice because it does not require the baseline hazard function and preserves promising statistical properties (Andersen and Gill, 1982; Tsiatis, 1981). Particularly, the partial likelihood pl_n can be obtained from the full likelihood by profiling out the baseline hazard (Van der Vaart, 1998):

$$\text{pl}_n(\eta_w(x)) = \sup_{\{\Lambda_i\}} \ell_n(\eta_w(x), \{\Lambda_i\}), \quad w \in \{0, 1\},$$

where Λ_i denotes the cumulating hazard at Y_i^c if the subject i is not censored. Let $\hat{\Lambda}_i$ be the baseline hazard estimators associated with the partial likelihood maximization, and the partial likelihood maximizer is equivalent to that of the full likelihood with $\hat{\Lambda}_i$. The connection motivates the method based on partial likelihood, which corresponds to the standard partial likelihood maximization method under the specification (18) (details are provided in Section 4 of the Supplementary Materials).

Proposition 2: *Under the conditions of Proposition 1, Assumption 1, Assumption 2, and assume no treatment effect,* $\|a_n(x) - a(x)\|_2$, $\|\nu_n(x) - \nu(x)\|_2 = O(c_n)$, $c_n \to 0$, *then*

$$\|\beta_n - \beta\|_2 = \tilde{O}\left(c_n^2 + n^{-1/2}\right).$$

For non-zero treatment effects, Proposition 2 is, in general, not true. Despite lacking theoretical guarantee, the partial-likelihood based method is empirically promising.

5 SIMULATION

5.1 Exponential family

We compare the following five meta-algorithms under the assumption $\tau(x) = x^\top\beta$. For nuisance function estimators,

Downloaded from https://academic.oup.com/biometrics/article/81/4/ujaf162/8403946 by guest on 07 June 2026

we obtain $\hat{e}(x)$ by logistic regression and $\hat{\eta}_0(x)$, $\hat{\eta}_1(x)$ by fitting generalized linear models. Results of estimating $\hat{\eta}_0(x)$, $\hat{\eta}_1(x)$ by tree boosting are available in the Supplementary Materials Section 5:

(1) Separate estimation (SE-learner). The separate estimation method estimates the control and treatment group mean functions $\hat{\eta}_0(x)$, $\hat{\eta}_1(x)$, takes the difference $\hat{\eta}_1(x) - \hat{\eta}_0(x)$, and further regresses the difference on the covariates to obtain $\hat{\beta}$.
(2) X-learner (X-learner).
X-learner first estimates the control group mean function $\hat{\eta}_0(x)$, and then estimates $\hat{\beta}$ by solving a generalized linear model with $\hat{\eta}_0(x)$ as the offset.
(3) Propensity score adjusted X-learner (PA-X-learner). Motivated by a thread of works (Dorie et al., 2019; Hahn et al., 2020; Vansteelandt and Daniel, 2014), we consider an augmented X-learner where the control group mean function is learned as a function of raw covariates, an estimated propensity score, and the interaction between.
(4) Direct extension of the R-learner (E-learner). The direct extension of the R-learner considers $a(x) = e(x)$ and the associated baseline $m(x) = \{1 - e(x)\}\eta_0(x) + e(x)\eta_1(x)$
for arbitrary response types (Dandl et al., 2024). The rest is the same as Algorithm 1.
(5) The proposed method (DINA-learner). We apply Algorithm 1 with the $a(x)$, $\nu(x)$ in (12). For Gaussian responses, the proposed method and the R-learner are the same; for other distributions, the two are different.

As for data generating mechanism, we consider $d = 5$ covariates independently generated from uniform $[-1, 1]$. The treatment assignment follows a logistic regression model $e(x) = e^{x^\top\theta}/\{1 + e^{x^\top\theta}\}$. The responses are sampled from the exponential family with natural parameter functions

$$\eta_0(x) = x^\top\alpha + \delta x_1 x_2, \quad \eta_1(x) = x^\top(\alpha + \beta) + \delta x_1 x_2, \tag{19}$$

for some $\delta \neq 0$. In both treatment and control groups, the response models are misspecified generalized linear models, while the difference of the natural parameters $\tau(x)$ is always linear. We consider continuous, binary, and discrete responses generated from Gaussian, Bernoulli, and Poisson distributions, respectively. For Cox model, we follow model (19) and use the baseline hazard function $\lambda(y) = y$, uniform censoring (75% units censored). We compare the five methods above based on partial-likelihood. We provide specific configurations of hyperparameters α, β, θ, δ in the Supplementary Materials Section 5.

We measure the estimation performance by the mean squared error $\mathbb{E}[\{\hat{\tau}(X) - \tau(X)\}^2]$, where the expectation is taken over the covariate population distribution. Results are summarized in Figure 2. Across three types of responses, our proposed method (DINA), direct extension of the R-learner (E), and propensity score adjusted X-learner (PA-X) performs relatively better than X-learner (X) and separate estimation (SE). Among the three well-performed methods, our proposed method approximately achieves the parametric convergence rate $O(n^{-1/2})$ and is more favorable for count data. As for X-learner and separate estimation, the errors stop decreasing as the sample size increases due to the bias.

In the Supplementary Materials Section 5, we employ bootstrap methods to construct confidence intervals for $\hat{\beta}$. The confidence intervals based on our estimator achieve the best coverage in average with comparable widths across settings.

6 REAL DATA ANALYSIS

We apply our method to the SPRINT dataset (Powers et al., 2018). The data are collected from a randomized trial containing 7517 units after preprocessing. The response is whether any of the major CVD events occur to a participant, and we use seven lab measurements as covariates (details of the dataset and its preprocessing are included in Section 5 of the Supplementary Materials). Our goal is to assess the heterogeneous treatment effect on the log-odds scale.

We consider the five methods in Section 5. As Table 1 shows, the proposed method, propensity score adjusted X-learner, and X-learner produce negative intercepts significant at the 5% level, which implies that for a patient with average covariate values, the treatment decreases the odds of experiencing any CVD events. As for heterogeneity in the treatment effect, our proposal finds EGFR as a significant effect modifier. In particular, the estimated coefficient of EGFR is negative, indicating that the treatment is more beneficial to patients with high EGFR. In Figure 3, we plot the average log-odds ratio in subgroups determined by subjects' EGFR values, which is approximately linear.

In the Supplementary Materials Section 5, we assess methods' sensitivity to treatment assignment mechanisms.

7 DISCUSSION

The heterogeneity of treatment effects is highly dependent on the measurement scale and choosing an appropriate scale is important as it can influence the analysis and interpretation of treatment effects. In this paper, we propose to quantify the treatment effect by DINA for general responses motivated by the exponential family and Cox model, in contrast to CATE. For non-continuous responses, e.g., binary and survival data, DINA is of more practical interest, e.g., relative risk and hazard ratio, and is convenient for modeling the influence of covariates to the treatment effects. In addition, we introduce a DINA estimator extending the R-learner for CATE, which is insensitive to nuisance function estimation and allows practitioners to use powerful off-the-

Downloaded from https://academic.oup.com/biometrics/article/81/4/ujaf162/8403946 by guest on 07 June 2026

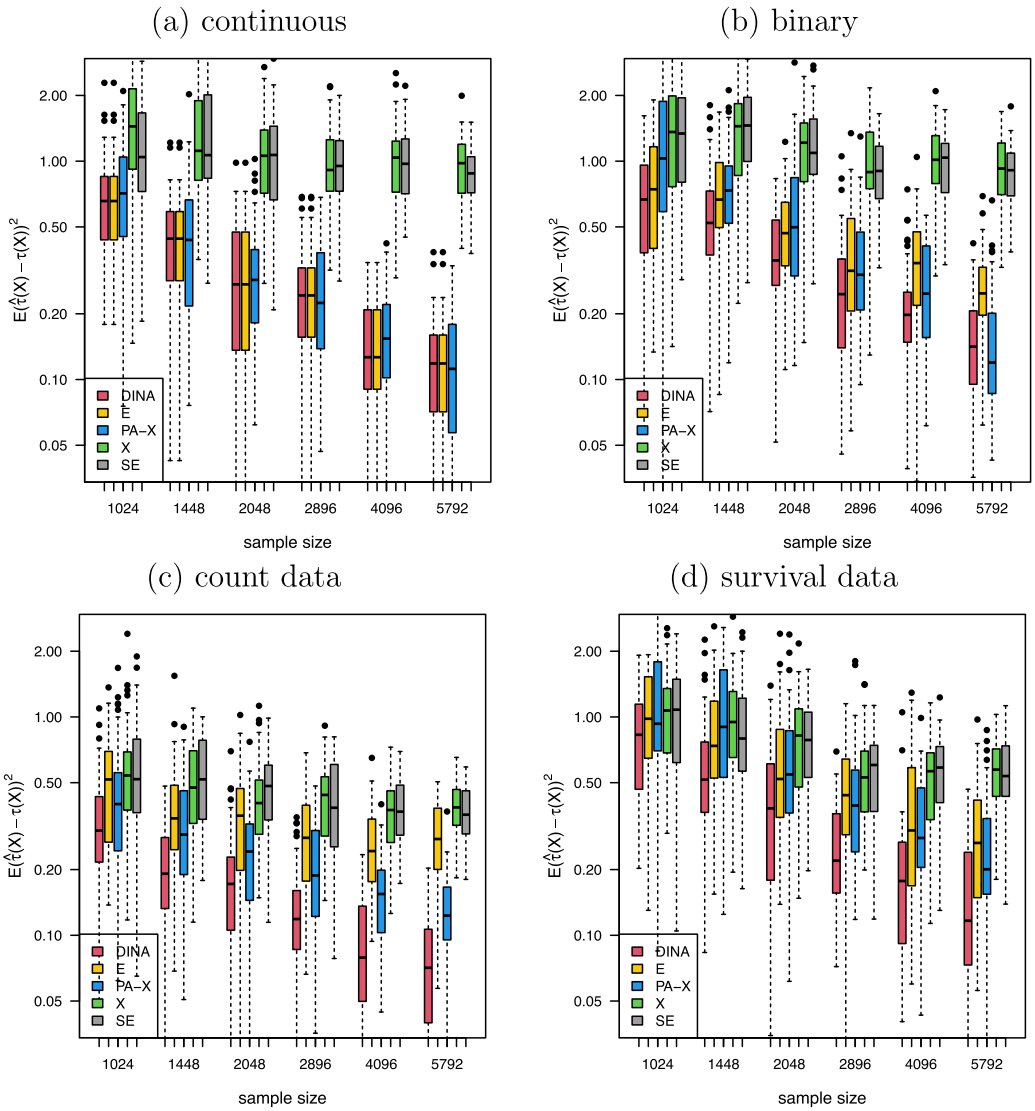


FIGURE 2 Estimation error log-log boxplots. We display the estimation errors $\mathbb{E}[(\hat{\tau}(X) - \tau(X))^2]$ over sample sizes in $[1024, 1448, 2048, 2896, 4096, 5792]$. We compare five methods: separate estimation (SE), X-learner (X), propensity score adjusted X-learner (PA-X), direct extension of the R-learner (E), and Algorithm 1 (DINA). We adopt the response model (19) and consider four types of responses: (a) continuous (Gaussian), (b) binary (Bernoulli), (c) count data (Poisson), (d) survival data (Cox model with uniform censoring, 75% units censored). We estimate the propensity score by logistic regression, and estimate the natural parameters by fitting generalized linear models or maximizing Cox partial likelihoods. We repeat all experiments 100 times. Abbreviation: DINA, DIfference in NAtural parameters.

shelf machine learning tools for nuisance-function estimation.

Our proposed method can be directly extended to quasi-likelihoods (Wedderburn, 1974) used to correct for over-dispersion of exponential family. For quasi-likelihood methods, it is not necessary to specify the full distribution of the outcome; instead, one only needs to define the relationship between the mean and the variance. In specific, let μ denote the mean and $V(\mu)$ denote the pre-determined variance function. The correspondence of natural parameter is defined through the derivative $d\eta = d\mu/V(\mu)$. The definition of nuisance functions (12) and Algorithm 1 still hold.

ACKNOWLEDGMENTS

We acknowledge the use of AI-based language tools to assist with the polishing and editing of the manuscript's text. All substantive content, ideas, and interpretations are solely those of the authors.

SUPPLEMENTARY MATERIALS

Supplementary material is available at *Biometrics* online.

Web Appendices, Algorithms, Proofs, Tables, and Figures referenced in Section 3-6, along with codes, are available with this paper at the Biometrics website on Oxford

Downloaded from https://academic.oup.com/biometrics/article/81/4/ujaf162/8403946 by guest on 07 June 2026

TABLE 1 Estimated coefficients and 95% CI from the SPRINT dataset.

Abbreviation		DINA	E	PA-X	X	SE
SBP	$\hat{\beta}_{\text{SBP}}$	0.109	0.101	0.313	0.255	0.135
	95% CI	[−0.169, 0.387]	[−0.166, 0.368]	[−0.030, 0.656]	[−0.072, 0.582]	[−0.128, 0.398]
EGFR	$\hat{\beta}_{\text{EGFR}}$	−0.331	−0.353	−0.107	−0.086	−0.087
	95% CI	[−0.662, 0.000]	[−0.725, 0.019]	[−0.548, 0.334]	[−0.412, 0.239]	[−0.432, 0.258]
CHR	$\hat{\beta}_{\text{CHR}}$	0.111	0.129	−0.176	−0.141	−0.121
	95% CI	[−0.273, 0.495]	[−0.238, 0.496]	[−0.552, 0.200]	[−0.480, 0.198]	[−0.491, 0.249]
GLUR	$\hat{\beta}_{\text{GLUR}}$	−0.211	−0.205	−0.116	−0.0535	−0.156
	95% CI	[−0.501, 0.079]	[−0.468, 0.058]	[−0.455, 0.223]	[−0.306, 0.199]	[−0.454, 0.142]
HDL	$\hat{\beta}_{\text{HDL}}$	−0.000	0.038	0.011	−0.030	−0.136
	95% CI	[−0.402, 0.401]	[−0.366, 0.442]	[−0.448, 0.470]	[−0.426, 0.366]	[−0.565, 0.293]
TRR	$\hat{\beta}_{\text{TRR}}$	0.103	0.098	0.218	0.082	0.026
	95% CI	[−0.267, 0.473]	[−0.262, 0.459]	[−0.219, 0.655]	[−0.230, 0.393]	[−0.305, 0.357]
UMALCR	$\hat{\beta}_{\text{UMALCR}}$	0.001	−0.045	−0.039	0.094	−0.057
	95% CI	[−0.228, 0.230]	[−0.284, 0.194]	[−0.257, 0.178]	[−0.124, 0.311]	[−0.253, 0.139]
Intercept	$\hat{\beta}_0$	−0.433	−0.412	−0.350	−0.336	−0.246
	95% CI	[−0.735, −0.131]	[−0.739, −0.085]	[−0.730, 0.030]	[−0.669, −0.003]	[−0.583, 0.091]

We compare the five methods in Section 5. We input the true propensity score $e(x) = 0.5$ and use random forests as the nuisance-function learner. We use bootstrap (100 bootstrap samples) to construct the confidence intervals.Abbreviations: CI, confidence intervals; DINA, DIfference in NAtural parameters.

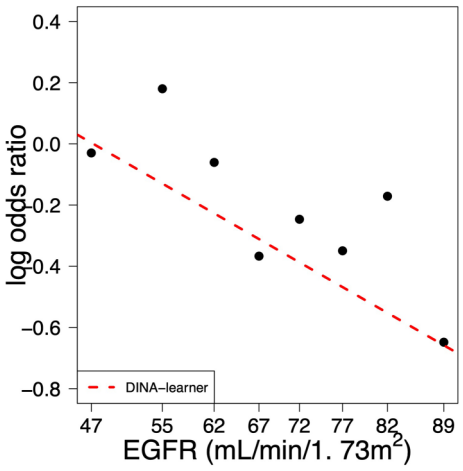


FIGURE 3 EGFR and DINA (log-odds ratio). We divide the participants into eight groups according to their EGFR. We plot the average log-odds ratio of the intervention. We also superimpose the estimated line from the DINA-learner, where we fix the values of the covariates other than EGFR at their averages. Abbreviation: DINA, DIfference in NAtural parameters. .

Academic. Codes are also accessible from https://github.com/ZijunGao/DINA.

FUNDING

This research was partially supported by grants DMS 2013736 and IIS 1837931 from the National Science Foundation, and grant 5R01 EB 001988-21 from the National Institutes of Health.

CONFLICT OF INTEREST

None declared.

DATA AVAILABILITY

The real dataset that support the findings in this paper can be provided upon request to the corresponding author.

REFERENCES

Andersen, P. K. and Gill, R. D. (1982). Cox's regression model for counting processes: a large-sample study. *The Annals of Statistics*, 10, 1100–1120.

Bennett, J. and Lanning, S. (2007). The Netflix prize. In *bibinfobtitleProceedings of the KDD Cup Workshop*, vol. 2007, pp. 3–6.San Jose, CA, USA.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W. et al. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21, C1–C68.

Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society: Series B (Methodological)*, 34, 187–202.

Dandl, S., Bender, A. and Hothorn, T. (2024). Heterogeneous treatment effect estimation for observational data using model-based forests. *Statistical Methods in Medical Research*, 33, 392–413.

Dewan, Y., Komolafe, E. O., Mejía-Mantilla, J. H., Perel, P., Roberts, I. and Shakur, H. (2012). CRASH-3-tranexamic acid for the treatment of significant traumatic brain injury: study protocol for an international randomized, double-blind, placebo-controlled trial. *Trials*, 13, 1–14.

Dorie, V., Hill, J., Shalit, U., Scott, M. and Cervone, D. (2019). Automated versus do-it-yourself methods for causal inference: Lessons learned from a data analysis competition. *Statistical Science*, 34, 43–68.

Glas, A. S., Lijmer, J. G., Prins, M. H., Bonsel, G. J. and Bossuyt, P. M. (2003). The diagnostic odds ratio: a single indicator of test performance. *Journal of Clinical Epidemiology*, 56, 1129–1135.

Hahn, P. R., Murray, J. S. and Carvalho, C. M. (2020). Bayesian regression tree models for causal inference: regularization, confounding, and heterogeneous effects (with discussion). *Bayesian Analysis*, 15, 965–1056.

Downloaded from https://academic.oup.com/biometrics/article/81/4/ujaf162/8403946 by guest on 07 June 2026

Imbens, G. W. and Rubin, D. B. (2015). *Causal Inference for Statistics, Social, and Biomedical Sciences*. Cambridge University Press.

Lesko, L. (2007). Personalized medicine: elusive dream or imminent reality? *Clinical Pharmacology and Therapeutics*, 81, 807–816.

Low, Y. S., Gallego, B. and Shah, N. H. (2016). Comparing high-dimensional confounder control methods for rapid cohort studies from electronic health records. *Journal of Comparative Effectiveness Research*, 5, 179–192.

Murphy, M., Redding, S. and Twyman, J. (2016). *Handbook on Personalized Learning for States, Districts, and Schools*. Philadelphia, PA: Center for Innovations in Learning.

Newey, W. K. (1994). The asymptotic variance of semiparametric estimators. *Econometrica: Journal of the Econometric Society*, 62, 1349–1382.

Nie, X. and Wager, S. (2021). Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika*, 108, 299–319.

Powers, S., Qian, J., Jung, K., Schuler, A., Shah, N. H., Hastie, T. et al. (2018). Some methods for heterogeneous treatment effect estimation in high dimensions. *Statistics in Medicine*, 37, 1767–1787.

Robinson, P. M. (1988). Root-n-consistent semiparametric regression. *Econometrica: Journal of the Econometric Society*, 56, 931–954.

Rosenbaum, P. R. et al. (2010). *Design of Observational Studies*. Springer.

Rothman, K. J. (2012). *Epidemiology: An Introduction*. Oxford: Oxford University Press.

Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66, 688–701.

Tian, L., Alizadeh, A. A., Gentles, A. J. and Tibshirani, R. (2014). A simple method for estimating interactions between a treatment and a large number of covariates. *Journal of the American Statistical Association*, 109, 1517–1532.

Tsiatis, A. A. (1981). A large-sample study of Cox's regression model. *The Annals of Statistics*, 9, 93–108.

Van der Vaart, A. (1998). *Asymptotic Statistics*. Cambridge: Cambridge University Press.

Vansteelandt, S. and Daniel, R. M. (2014). On regression adjustment for the propensity score. *Statistics in Medicine*, 33, 4053–4072.

Wedderburn, R. W. (1974). Quasi-likelihood functions, generalized linear models, and the Gauss—Newton method. *Biometrika*, 61, 439–447.

Wendling, T., Jung, K., Callahan, A., Schuler, A., Shah, N. and Gallego, B. (2018). Comparing methods for estimation of heterogeneous treatment effects using observational data from health care databases. *Statistics in Medicine*, 37, 3309–3324.

Downloaded from https://academic.oup.com/biometrics/article/81/4/ujaf162/8403946 by guest on 07 June 2026

Received: June 28, 2024; Revised: September 27, 2025; Accepted: October 9, 2025
© The Author(s) 2025. Published by Oxford University Press on behalf of The International Biometric Society. This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (https://creativecommons.org/licenses/by-nc/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site-for further information please contact journals.permissions@oup.com